

Lovely Professional University, Punjab

Course Code	Course Title	Lectures	Tutorials	Practicals	Credits	
CHE110	ENVIRONMENTAL STUDIES	2	0	0	2	
Course Weightage	ATT: 5 CA: 40 MTT: 20 ETT: 35	Exam Category: 11: Mid Term Exam: All MCQ – End Term Exam: All MCQ				
Course Focus	EMPLOYABILITY,SKILL DEVELOPMENT					

Course Outcomes :Through this course students should be able to

CO1 :: define current environmental issues and associated problems.

CO2 :: explain various components of environment and their role in maintaining the ecosystem and biodiversity.

CO3 :: determine the role of various environmental policies, practices and techniques to reduce environmental pollution.

CO4 :: analyze various environmental phenomena and their relationship with human communities.

	TextBooks (T)		
Sr No	Title	Author	Publisher Name
T-1	PERSPECTIVE IN ENVIRONMENTAL STUDIES	ANUBHA KAUSHIK, C P KAUSHIK	NEW AGE INTERNATIONAL PUBLISHERS
	Reference Books (R)		
Sr No	Title	Author	Publisher Name
R-1	TEXT BOOK OF ENVIRONMENTAL STUDIES	D. DAVE AND S. S. KATEWA	CENGAGE LEARNING

Other Reading (OR)

Sr No	Journals articles as Compulsary reading (specific articles, complete reference)
OR-1	https://www.ugc.gov.in/oldpdf/modelcurriculum/env.pdf , ,
OR-2	https://nitsri.ac.in/Department/CHEMISTRY/EVS_MATERIAL_2.pdf , ,
OR-3	https://www.youtube.com/watch?v=CebTVhzigOA ,

Relevant Websites (RW)

Sr No	(Web address) (only if relevant to the course)	Salient Features
RW-1	http://www.nei.org/master-document-folder/backgrounders/fact-sheets/chernobyl-accident-and-its-consequences	Chernobyl Accident
RW-2	https://www.cbd.int/convention/	Convention on Biological Diversity

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RW-3	https://www.bhopal.org/continuing-disaster/the-bhopal-gas-disaster/union-carbides-disaster/	Bhopal Gas Tragedy
RW-4	https://nmcg.nic.in/pollution.aspx	Ganga pollution
RW-5	http://pdf.usaid.gov/pdf_docs/PNAAT747.pdf	Silent Valley
RW-6	https://ecologise.in/2017/05/28/the-bishnois-indias-original-environmentalists-who-inspired-the-chipko-movement/	Bishnois of Rajasthan
RW-7	http://www.newagepublishers.com/samplechapter/001426.pdf	Multidisciplinary nature of environment
RW-8	http://www.newagepublishers.com/samplechapter/000964.pdf	Basics of Environmental studies
RW-9	https://www.iisd.org/mission-and-goals/sustainable-development	Sustainable Development
RW-10	https://www.un.org/en/academic-impact/sustainability	Sustainability
RW-11	https://sdgs.un.org/goals	Sustainable development goals
RW-12	https://scdhec.gov/sites/default/files/Library/OR-0689.pdf	Natural resource
RW-13	http://www.un-documents.net/ocf-07.htm	Non-renewable resource
RW-14	http://www.oas.org/dsd/publications/Unit/oea79e/ch05.htm	Renewable resource
RW-15	https://www.thegef.org/what-we-do/topics/land-degradation	Land degradation
RW-16	https://fsi.nic.in/uploads/isfr2023/isfr_book_eng-vol-1_2023.pdf	Forest cover of India
RW-17	https://www.unep.org/topics/forests/why-do-forests-matter	Importance of Forest Resources
RW-18	https://www.un.org/en/global-issues/water	Water resources and issues
RW-19	https://www.nationalgeographic.org/encyclopedia/ecosystem/	Ecosystem
RW-20	https://www.biologyonline.com/dictionary/ecological-pyramid	Ecological pyramids
RW-21	https://news.uchicago.edu/explainer/what-is-ecological-succession	Ecological Succession
RW-22	https://www.esa.org/wp-content/uploads/2012/12/biodiversity.pdf	Levels of Biodiversity
RW-23	https://www.conservation.org/priorities/biodiversity-hotspots	Global Biodiversity Hot Spots
RW-24	https://www.conservation.org/priorities/biodiversity-hotspots	Biodiversity hot spot
RW-25	http://pdf.usaid.gov/pdf_docs/PNABW370.pdf	Modes of Biodiversity conservation
RW-26	https://www.nrdc.org/stories/water-pollution-everything-you-need-know	Water pollutants, sources of water pollution
RW-27	https://oceanservice.noaa.gov/facts/eutrophication.html	Eutrophication
RW-28	https://programs.wcs.org/india/Newsroom/Blog/ID/15443/Biomagnification-An-Ever-growing-Threat	Biomagnification

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RW-29	http://www.yourownhealthandfitness.org/Documents/Hagler_Noise_pollution.pdf	Sources, effects of noise pollution, control measures of noise pollution
RW-30	http://www.envirocomp.org/books/chapters/1aap.pdf	Primary and Secondary pollutants
RW-31	https://www.conserve-energy-future.com/radioactive-pollution-causes-effects-solutions.php	Nuclear Hazards (Effects & Control)
RW-32	https://www.ecology.edu/environmental-legislation.html	Introduction to Environmental Legislation in India
RW-33	https://www.nrdc.org/stories/air-pollution-everything-you-need-know	Global warming, Smog, Control measures for air pollution
RW-34	http://idosi.org/wasj/wasj7%281%29/10.pdf	Green house effect, Acid rain, Depletion of ozone layer
RW-35	https://www.ugc.gov.in/oldpdf/modelcurriculum/Chapter7.pdf	Human population and environment
RW-36	https://ndma.gov.in/	Disaster Management in India

Audio Visual Aids (AV)

Sr No	(AV aids) (only if relevant to the course)	Salient Features
AV-1	https://www.youtube.com/watch?v=7G3eXI_DPn8	Introduction to Environmental Science
AV-2	https://www.youtube.com/watch?v=Sk5ELLPsD80	Introduction to Environmental Science
AV-3	https://www.youtube.com/watch?v=Sk5ELLPsD80	Environmental Science and Sustainability
AV-4	https://www.youtube.com/watch?v=wwP7474y8Jw	Sustainable Development
AV-5	https://www.youtube.com/watch?v=eEFwaQej_0E	Sustainable Development
AV-6	http://study.com/academy/lesson/what-are-natural-resources-definition-lesson-quiz.html	Classification of natural resources
AV-7	http://study.com/academy/lesson/soil-profile-definition-development-types.html	Soil profile
AV-8	https://www.youtube.com/watch?v=iMy5-Npr69E	Land degradation
AV-9	https://www.youtube.com/watch?v=gRJBuM7qjQ0	Desertification
AV-10	https://www.youtube.com/watch?v=a0S8iayJDhQ	Causes and impacts due to mining
AV-11	https://www.youtube.com/watch?v=m4WBbSv_N7U	Sources of water
AV-12	https://pmm.nasa.gov/education/videos/water-cycle-animation	Water cycle
AV-13	https://www.youtube.com/watch?v=kxqbpPWTl6A	Importance of water
AV-14	https://www.youtube.com/watch?v=_v8MEaejTok	Conflicts over water
AV-15	https://www.youtube.com/watch?v=Xo1Jyzba7rA	Droughts
AV-16	http://study.com/academy/lesson/what-are-floods-causes-types-prevention.html	Floods
AV-17	https://www.youtube.com/watch?v=QXiHSgOEL_M	Over-exploitation of surface and ground water
AV-18	https://www.youtube.com/watch?v=5lK_fs3p7yc	Use of surface and ground water
AV-19	https://www.youtube.com/watch?v=_dWJVHIE9S8	Importance of forests

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AV-20	http://study.com/academy/lesson/what-is-sustainable-forest-management-definition-and-examples.html	Sustainable forest management
AV-21	http://study.com/academy/lesson/renewable-non-renewable-resources-definition-differences.html	Non-renewable energy sources
AV-22	http://study.com/academy/lesson/what-is-a-renewable-energy-source-definition-example-quiz.html	Renewable Energy Source
AV-23	https://www.youtube.com/watch?v=E6dX1i9s1I0	Use of alternate energy sources
AV-24	https://www.youtube.com/watch?v=f_x_saJEGiw	Energy recovery
AV-25	https://www.youtube.com/watch?v=xP0MuoYnMI0	Case studies: India's-rising-energy-needs
AV-26	https://www.youtube.com/watch?v=auBFn-u9b6A	JK Energy Solutions Case Study
AV-27	https://www.youtube.com/watch?v=Da6DhVAeN9s	Case Studies in Municipal Energy Conservation
AV-28	http://study.com/academy/lesson/food-chains-trophic-levels-and-energy-flow-in-an-ecosystem.html	Food chains
AV-29	https://www.youtube.com/watch?v=gB4UDHCvXMU	Grassland Ecosystem
AV-30	http://study.com/academy/lesson/aquatic-ecosystems-abiotic-factors.html	Aquatic Ecosystems
AV-31	https://www.youtube.com/watch?v=f8eCbwg-lik	Forest Ecosystem
AV-32	https://www.youtube.com/watch?v=Nbi4wW7ojXs	Desert Ecosystem
AV-33	http://study.com/academy/lesson/what-is-ecological-succession-definition-types-stages.html	Ecological succession
AV-34	https://www.youtube.com/watch?v=GK_vRtHJZu4	Biodiversity
AV-35	https://www.youtube.com/watch?v=7tgNamjTRkk	What is biodiversity and why is it important
AV-36	https://www.youtube.com/watch?v=Jtuh368CW3c	Levels of biological diversity: genetic, species and ecosystem diversity
AV-37	https://www.youtube.com/watch?v=VoS1-yOeEo4	Values of biodiversity
AV-38	https://www.youtube.com/watch?v=zJyxOYfvIfA	Threats to biodiversity: Habitat loss
AV-39	https://www.youtube.com/watch?v=GK_vRtHJZu4	India as a mega diversity nation
AV-40	https://www.youtube.com/watch?v=JZv3LOMiI9Q	Endangered and endemic species of India
AV-41	https://www.youtube.com/watch?v=bTsxx1KZwlM	Water pollution
AV-42	https://www.youtube.com/watch?v=00c4goQRLek	Soil Pollution
AV-43	https://www.youtube.com/watch?v=yhWnrNQPwAs	Solid waste management: Control measures of urban and industrial waste
AV-44	https://www.youtube.com/watch?v=3nx5yr2GnGk	Ill-effects of fireworks
AV-45	https://www.youtube.com/watch?v=6UoUUkaMyl0	Nuclear hazards. Cause, effect and control
AV-46	https://www.youtube.com/watch?v=KxlgRHf_7oA	Radioactive Pollution
AV-47	https://www.youtube.com/watch?v=-VaaTn_g1_k	Noise Pollution
AV-48	https://www.youtube.com/watch?v=nnBuQe_NKhE	Acid rain and impacts on human communities and agriculture

AV-49	https://www.youtube.com/watch?v=1sQbz1IZ1Ys	Ozone layer depletion
AV-50	https://www.youtube.com/watch?v=CTUOchYZG2k	Environment Laws
AV-51	https://www.youtube.com/watch?v=tYJOd-QYidc	Wild life conservation act
AV-52	https://www.youtube.com/watch?v=6U_4jX4engM	Forest conservation act
AV-53	https://www.youtube.com/watch?v=hPgBLOZ85Cw	Environment protection act
AV-54	https://www.youtube.com/watch?v=p1sG3DX-bII	Montreal Protocol Documentary
AV-55	https://www.youtube.com/watch?v=ThQiJpbLpp0&pp=ygUYQmFzaWNzIG9mIEt5b3RvIFByb3RvY29s	Kyoto Protocol 1997
AV-56	https://www.youtube.com/watch?v=fTznEIZRkLg	Population growth
AV-57	http://study.com/academy/lesson/comparing-life-centered-human-centeredenvironmental-ethics.html	Environmental ethics
AV-58	https://www.youtube.com/watch?v=YIrKW6jXjdM	Environmental Conservation

LTP week distribution: (LTP Weeks)	
Weeks before MTE	7
Weeks After MTE	7
Spill Over (Lecture)	4

Detailed Plan For Lectures

Week Number	Lecture Number	Broad Topic(Sub Topic)	Chapters/Sections of Text/reference books	Other Readings, Relevant Websites, Audio Visual Aids, software and Virtual Labs	Lecture Description	Learning Outcomes	Pedagogical Tool Demonstration/ Case Study / Images / animation / ppt etc. Planned	Live Examples

Week 1	Lecture 1	Introduction and sustainable development(introduction to environment)	T-1 R-1	OR-1 RW-7 RW-8 AV-1	Zero lecture for the introduction to the course objectives, structure and details of academic tasks. Introduction to environment and its various components. Multidisciplinary nature, scope and importance of environmental studies.	Understand the course objectives and academic details. Demonstrate global empathy for saving the world from environmental destruction. Recognize the interconnections between environmental studies and other subject.	Lecture delivery with discussion.	Medha Patkar, Maneka Gandhi, Anil Kumar Agarwal, Sunita narain, M S Swaminathan, Chandra Bhushan and many other renowned dignitaries and personages who have been treasured by distinguished awards in various disciplines now directing and managing centre for science and environment-India top NGO to analyses and study the relationship between environment and development and create public consciousness about the need for sustainable development.
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Week 1	Lecture 1	Introduction and sustainable development(spheres of earth)	T-1 R-1	RW-7 RW-8 AV-1	Zero lecture for the introduction to the course objectives, structure and details of academic tasks. Introduction to environment and its various components. Multidisciplinary nature, scope and importance of environmental studies.	Understand the course objectives and academic details. Demonstrate global empathy for saving the world from environmental destruction. Recognize the interconnections between environmental studies and other subject.	Lecture delivery with discussion.	Medha Patkar, Maneka Gandhi, Anil Kumar Agarwal, Sunita narain, M S Swaminathan, Chandra Bhushan and many other renowned dignitaries and personages who have been treasured by distinguished awards in various disciplines now directing and managing centre for science and environment-India top NGO to analyses and study the relationship between environment and development and create public consciousness about the need for sustainable development.
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Week 1	Lecture 2	Introduction and sustainable development(introduction to environment)	T-1 R-1	OR-1 RW-7 RW-8 AV-1	Zero lecture for the introduction to the course objectives, structure and details of academic tasks. Introduction to environment and its various components. Multidisciplinary nature, scope and importance of environmental studies.	Understand the course objectives and academic details. Demonstrate global empathy for saving the world from environmental destruction. Recognize the interconnections between environmental studies and other subject.	Lecture delivery with discussion.	Medha Patkar, Maneka Gandhi, Anil Kumar Agarwal, Sunita narain, M S Swaminathan, Chandra Bhushan and many other renowned dignitaries and personages who have been treasured by distinguished awards in various disciplines now directing and managing centre for science and environment-India top NGO to analyses and study the relationship between environment and development and create public consciousness about the need for sustainable development.
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Week 1	Lecture 2	Introduction and sustainable development(spheres of earth)	T-1 R-1	RW-7 RW-8 AV-1	Zero lecture for the introduction to the course objectives, structure and details of academic tasks. Introduction to environment and its various components. Multidisciplinary nature, scope and importance of environmental studies.	Understand the course objectives and academic details. Demonstrate global empathy for saving the world from environmental destruction. Recognize the interconnections between environmental studies and other subject.	Lecture delivery with discussion.	Medha Patkar, Maneka Gandhi, Anil Kumar Agarwal, Sunita narain, M S Swaminathan, Chandra Bhushan and many other renowned dignitaries and personages who have been treasured by distinguished awards in various disciplines now directing and managing centre for science and environment-India top NGO to analyses and study the relationship between environment and development and create public consciousness about the need for sustainable development.
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Week 2	Lecture 3	Introduction and sustainable development(concept of sustainability)	T-1 R-1	RW-9 RW-10 AV-3 AV-4	sustainable development concept of sustainability carrying capacity pillars of sustainability	Students will understand the concepts and importance of sustainability and pillars of sustainable development.	Lecture delivery with ppt and discussion.	Reducing your plastic use can have all kinds of positive effects for the planet—from reducing waste to conserving energy and resources. Eliminate plastic bags and bring a reusable tote to stores to lessen plastic waste. Use of modern irrigation water to conserve water in agriculture/industrial sector.
		Introduction and sustainable development(sustainable development)	T-1 R-1	RW-9 RW-10 AV-3 AV-4	sustainable development concept of sustainability carrying capacity pillars of sustainability	Students will understand the concepts and importance of sustainability and pillars of sustainable development.	Lecture delivery with ppt and discussion.	Reducing your plastic use can have all kinds of positive effects for the planet—from reducing waste to conserving energy and resources. Eliminate plastic bags and bring a reusable tote to stores to lessen plastic waste. Use of modern irrigation water to conserve water in agriculture/industrial sector.

Week 2	Lecture 3	Introduction and sustainable development(carrying capacity)	T-1 R-1	RW-9 RW-10 AV-3 AV-4	sustainable development concept of sustainability carrying capacity pillars of sustainability	Students will understand the concepts and importance of sustainability and pillars of sustainable development.	Lecture delivery with ppt and discussion.	Reducing your plastic use can have all kinds of positive effects for the planet—from reducing waste to conserving energy and resources. Eliminate plastic bags and bring a reusable tote to stores to lessen plastic waste. Use of modern irrigation water to conserve water in agriculture/industrial sector.
		Introduction and sustainable development(pillars of sustainability)	T-1 R-1	RW-9 RW-10 AV-3 AV-4	sustainable development concept of sustainability carrying capacity pillars of sustainability	Students will understand the concepts and importance of sustainability and pillars of sustainable development.	Lecture delivery with ppt and discussion.	Reducing your plastic use can have all kinds of positive effects for the planet—from reducing waste to conserving energy and resources. Eliminate plastic bags and bring a reusable tote to stores to lessen plastic waste. Use of modern irrigation water to conserve water in agriculture/industrial sector.

Week 2	Lecture 4	Introduction and sustainable development(sustainable development goals)	T-1 R-1	OR-1 RW-9 RW-11 AV-5	Sustainable development goals	Student will learn about sustainable development goals, pathway for achieving SDG and progress of various countries in contribution to SDG in this lecture.	Lecture delivery with ppt and discussion	Clean water and sanitation Water scarcity affects more than 40% of people around the world, and that number is projected to increase due to climate change
Week 3	Lecture 5	Natural resources and ecosystem(introduction and classification)	T-1 R-1	RW-12 AV-6 AV-7 AV-8 AV-9 AV-10	Natural resources and classification Land resource, soil profile, function of soil. Cause, effect and control methods of land degradation, soil erosion and desertification. After the lecture delivery, last 15 minutes is used to allotment of the assignment/term paper topics.	Examine natural resources and land resources. Investigate soil erosion and its control methods. Explore the concept of desertification.	Lecture delivery with ppt and discussion.	The effect of desertification in Northern Gujarat and expansion of deserts across globe. Expansion of Sahara and Thar desert.
		Natural resources and ecosystem(renewable and non renewable resources)	T-1 R-1	RW-12 AV-6 AV-7 AV-8 AV-9 AV-10	Natural resources and classification Land resource, soil profile, function of soil. Cause, effect and control methods of land degradation, soil erosion and desertification. After the lecture delivery, last 15 minutes is used to allotment of the assignment/term paper topics.	Examine natural resources and land resources. Investigate soil erosion and its control methods. Explore the concept of desertification.	Lecture delivery with ppt and discussion.	The effect of desertification in Northern Gujarat and expansion of deserts across globe. Expansion of Sahara and Thar desert.
		Natural resources and ecosystem(land resources, land degradation, soil erosion and desertification)	T-1 R-1	RW-12 RW-15 AV-6 AV-7 AV-8 AV-9 AV-10	Natural resources and classification Land resource, soil profile, function of soil. Cause, effect and control methods of land degradation, soil erosion and desertification. After the lecture delivery, last 15 minutes is used to allotment of the assignment/term paper topics.	Examine natural resources and land resources. Investigate soil erosion and its control methods. Explore the concept of desertification.	Lecture delivery with ppt and discussion.	The effect of desertification in Northern Gujarat and expansion of deserts across globe. Expansion of Sahara and Thar desert.

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Week 3	Lecture 6	Natural resources and ecosystem(forest resources, causes and impacts of deforestation)	T-1 R-1	RW-16 RW-17 RW-18 AV-11 AV-12 AV-13 AV-14 AV-17 AV-18 AV-19 AV-20	Forest resource: Importance of forests: Direct and indirect benefits. Deforestation: causes, effect and control. Water resource: Importance of water, Source of water, water cycle. ,over-exploitation of ground water. Water shortage and conflicts over water.	Recognize the importance and benefits of forests. Analyze the concept of deforestation. Investigate water resources and related challenges concerning water availability and scarcity.	Lecture delivery with discussion and using videos related to deforestation.	Forest status of India and other countries and rate of deforestation across countries. Water stress countries, Drought in Kalahandi (Government induced drought), Save Ganga Movement, Ganga action plan. Eco-task force to check the Ganga pollution. Cherrapunji-Drinking the sky. Still unsettled Kaveri dispute
		Natural resources and ecosystem(water resources, use and over-exploitation of surface and ground water, conflicts over water)	T-1 R-1	RW-16 RW-17 RW-18 AV-19 AV-20	Forest resource: Importance of forests: Direct and indirect benefits. Deforestation: causes, effect and control. Water resource: Importance of water, Source of water, water cycle. ,over-exploitation of ground water. Water shortage and conflicts over water.	Recognize the importance and benefits of forests. Analyze the concept of deforestation. Investigate water resources and related challenges concerning water availability and scarcity.	Lecture delivery with discussion and using videos related to deforestation.	Forest status of India and other countries and rate of deforestation across countries. Water stress countries, Drought in Kalahandi (Government induced drought), Save Ganga Movement, Ganga action plan. Eco-task force to check the Ganga pollution. Cherrapunji-Drinking the sky. Still unsettled Kaveri dispute

Week 4	Lecture 7	Natural resources and ecosystem(energy resources, use of alternate energy sources, growing energy needs, conservation of natural resources and human role)	T-1 R-1	RW-13 RW-14 AV-21 AV-22 AV-23 AV-24 AV-25 AV-26 AV-27	Energy Resources: Renewable and non-renewable energy resources. Use of alternative energy resource and limitations of alternative energy resource. Energy recovery. Growing energy need and problems related to energy	Identify various renewable and non-renewable energy resources. Analyze the significance of alternative energy resources.	Identify various renewable and non-renewable energy resources. Analyze the significance of alternative energy resources. Lecture delivery with discussion and using videos related to various sources of energy.	Solar energy flow as the single energy source that is perpetual and causes the flow of water and air currents that are renewable resources. Think about Petroleum and coal reserves of the world which are getting depleted rampantly-Fossil fuel reserves are non-renewable. Current development in alternative sources of energy
	Lecture 8	Natural resources and ecosystem(ecosystem: structure and function of ecosystem)	T-1 R-1	RW-19 AV-28	Introduction and classification of ecosystem. Structure of ecosystem: biotic and abiotic components. Function of ecosystem: food chain, food web, flow of energy. Concept of trophic level	Understand the concept of ecosystems, including classification and the structure of biotic and abiotic components. Analyze ecosystems, including food chains, food webs, and energy flow within them.	Lecture delivery with videos, animations, and discussion.	Different types of ecosystem across world. Destruction of ecosystem of Sahara deserts. Destruction of ecosystem by taking example of Yamuna river. Food chain in different ecosystems.

Week 4	Lecture 8	Natural resources and ecosystem(types of ecosystem)	T-1 R-1	RW-19 AV-28	Introduction and classification of ecosystem. Structure of ecosystem: biotic and abiotic components. Function of ecosystem: food chain, food web, flow of energy. Concept of trophic level	Understand the concept of ecosystems, including classification and the structure of biotic and abiotic components. Analyze ecosystems, including food chains, food webs, and energy flow within them.	Lecture delivery with videos, animations, and discussion.	Different types of ecosystem across world. Destruction of ecosystem of Sahara deserts. Destruction of ecosystem by taking example of Yamuna river. Food chain in different ecosystems.
		Natural resources and ecosystem(energy flow in an ecosystem: food chains and food webs)	T-1 R-1	RW-19 AV-28	Introduction and classification of ecosystem. Structure of ecosystem: biotic and abiotic components. Function of ecosystem: food chain, food web, flow of energy. Concept of trophic level	Understand the concept of ecosystems, including classification and the structure of biotic and abiotic components. Analyze ecosystems, including food chains, food webs, and energy flow within them.	Lecture delivery with videos, animations, and discussion.	Different types of ecosystem across world. Destruction of ecosystem of Sahara deserts. Destruction of ecosystem by taking example of Yamuna river. Food chain in different ecosystems.
Week 5	Lecture 9	Natural resources and ecosystem(ecological pyramids)	T-1 R-1	RW-20	Concepts of ecological pyramids, Types of pyramid and shapes within pyramid of number, biomass and energy .	Student will understand and examine the structure and characteristics of ecological pyramids.	Lecture delivery with videos, animations, and discussion.	
	Lecture 10	Natural resources and ecosystem(ecological succession)	T-1 R-1	RW-21 AV-29 AV-30 AV-31 AV-32 AV-33	Ecological succession: Types and process of succession. Structure and function of Forest ecosystem, Grassland ecosystem, Desert ecosystem, Aquatic ecosystems: freshwater and marine water ecosystem.	Explore the process of ecological succession and structure and characteristics of various ecosystems on land and water	Lecture delivery with videos, animations, and discussion.	Rivers, lakes and ocean as distinct ecosystem.

Week 5	Lecture 10	Natural resources and ecosystem(Case studies of the following ecosystems : a)forest ecosystem b) grassland ecosystem c) desert ecosystem d) aquatic ecosystem)	T-1 R-1	RW-21 AV-29 AV-30 AV-31 AV-32 AV-33	Ecological succession: Types and process of succession. Structure and function of Forest ecosystem, Grassland ecosystem, Desert ecosystem, Aquatic ecosystems: freshwater and marine water ecosystem.	Explore the process of ecological succession and structure and characteristics of various ecosystems on land and water	Lecture delivery with videos, animations, and discussion.	Rivers, lakes and ocean as distinct ecosystem.
Week 6	Lecture 11	Biodiversity and conservation(definition)	T-1 R-1	RW-22 RW-23 RW-24 AV-34 AV-35 AV-36 AV-37	Introduction to biodiversity, types of biodiversity such as genetic, species and ecosystem diversity. Values of biodiversity. Biodiversity hot spots.	Investigate biodiversity and its different types. Raise awareness about the significance of biodiversity and biodiversity hotspots.	Lecture delivery with discussion and by showing videos.	Genetic diversity of rice in India, Advantages of biodiversity to developing countries. Endangered Bengal Tiger, Black buck, Indian Rhino. The western ghats and Eastern Himalayas region.
		Biodiversity and conservation(levels of biological diversity: genetic, species and ecosystem diversity)	T-1 R-1	RW-22 RW-23 RW-24 AV-34 AV-35 AV-36 AV-37	Introduction to biodiversity, types of biodiversity such as genetic, species and ecosystem diversity. Values of biodiversity. Biodiversity hot spots.	Investigate biodiversity and its different types. Raise awareness about the significance of biodiversity and biodiversity hotspots.	Lecture delivery with discussion and by showing videos.	Genetic diversity of rice in India, Advantages of biodiversity to developing countries. Endangered Bengal Tiger, Black buck, Indian Rhino. The western ghats and Eastern Himalayas region.

Week 6	Lecture 11	Biodiversity and conservation(importance of biodiversity:ecosystem and biodiversity services, ecological, economic, social, aesthetic and informational value)	T-1 R-1	RW-22 RW-23 RW-24 AV-34 AV-35 AV-36 AV-37	Introduction to biodiversity, types of biodiversity such as genetic, species and ecosystem diversity. Values of biodiversity. Biodiversity hot spots.	Investigate biodiversity and its different types. Raise awareness about the significance of biodiversity and biodiversity hotspots.	Lecture delivery with discussion and by showing videos.	Genetic diversity of rice in India, Advantages of biodiversity to developing countries. Endangered Bengal Tiger, Black buck, Indian Rhino. The western ghats and Eastern Himalayas region.
		Biodiversity and conservation(hot spots of biodiversity)	T-1 R-1	RW-22 RW-23 RW-24 AV-34 AV-35 AV-36 AV-37	Introduction to biodiversity, types of biodiversity such as genetic, species and ecosystem diversity. Values of biodiversity. Biodiversity hot spots.	Investigate biodiversity and its different types. Raise awareness about the significance of biodiversity and biodiversity hotspots.	Lecture delivery with discussion and by showing videos.	Genetic diversity of rice in India, Advantages of biodiversity to developing countries. Endangered Bengal Tiger, Black buck, Indian Rhino. The western ghats and Eastern Himalayas region.
	Lecture 12	Biodiversity and conservation(threats to biodiversity: habitat loss, poaching of wildlife, biological invasions)		AV-38 AV-40	Threats to biodiversity due to habitat destruction, poaching, pollution, climate change, biological invasion, legal system, mining. Categories of species on the basis of threat: endangered, vulnerable, threaten, edge, endemic and rare species.	Appraise the importance of biodiversity and its values. Analyze the primary reasons behind the destruction of biodiversity.	Lecture delivery with discussion and videos	Indian Sandal wood plant is a vulnerable species owing to exploitation and smuggling of wood for fine furniture. Destruction of Biodiversity in Tehri Dam Uttarakhand

Week 6	Lecture 12	Biodiversity and conservation(human wildlife conflicts)		AV-38 AV-40	Threats to biodiversity due to habitat destruction, poaching, pollution, climate change, biological invasion, legal system, mining. Categories of species on the basis of threat: endangered, vulnerable, threaten, edge, endemic and rare species.	Appraise the importance of biodiversity and its values. Analyze the primary reasons behind the destruction of biodiversity.	Lecture delivery with discussion and videos	Indian Sandal wood plant is a vulnerable species owing to exploitation and smuggling of wood for fine furniture. Destruction of Biodiversity in Tehri Dam Uttarakhand
		Biodiversity and conservation(endangered and endemic species in India)		AV-38 AV-40	Threats to biodiversity due to habitat destruction, poaching, pollution, climate change, biological invasion, legal system, mining. Categories of species on the basis of threat: endangered, vulnerable, threaten, edge, endemic and rare species.	Appraise the importance of biodiversity and its values. Analyze the primary reasons behind the destruction of biodiversity.	Lecture delivery with discussion and videos	Indian Sandal wood plant is a vulnerable species owing to exploitation and smuggling of wood for fine furniture. Destruction of Biodiversity in Tehri Dam Uttarakhand
Week 7	Lecture 13	Biodiversity and conservation(conservation of biodiversity: in-situ and ex-situ conservation methods of biodiversity)	T-1 R-1	RW-25 AV-39	Methods of conservation of biodiversity - In situ conservation: national parks, wildlife sanctuary and biosphere reserves. The merits and demerits of each of the conservation method and India as a mega bio diverse nation. The progress report of the field work based term paper will be collected at the end of this lecture.	Recognize the values of biodiversity and its significance. Explore various methods of biodiversity conservation and understanding India as mega bio diverse nation.	Lecture delivery with discussion and videos.	Project Tiger, Rhino and crocodile.

Week 7	Lecture 13	Biodiversity and conservation(biogeographic zones of India)	T-1 R-1	RW-25 AV-39	Methods of conservation of biodiversity - In situ conservation: national parks, wildlife sanctuary and biosphere reserves. The merits and demerits of each of the conservation method and India as a mega bio diverse nation. The progress report of the field work based term paper will be collected at the end of this lecture.	Recognize the values of biodiversity and its significance. Explore various methods of biodiversity conservation and understanding India as mega bio diverse nation.	Lecture delivery with discussion and videos.	Project Tiger, Rhino and crocodile.
		Biodiversity and conservation(India as a mega diversity nation)	T-1 R-1	RW-25 AV-39	Methods of conservation of biodiversity - In situ conservation: national parks, wildlife sanctuary and biosphere reserves. The merits and demerits of each of the conservation method and India as a mega bio diverse nation. The progress report of the field work based term paper will be collected at the end of this lecture.	Recognize the values of biodiversity and its significance. Explore various methods of biodiversity conservation and understanding India as mega bio diverse nation.	Lecture delivery with discussion and videos.	Project Tiger, Rhino and crocodile.
		SPILL OVER						
Week 7	Lecture 14				Spill Over			
		MID-TERM						

Week 8	Lecture 15	Environmental pollution (definition)	T-1 R-1	RW-26 RW-27 RW-28 RW-29 RW-30 AV-41 AV-42 AV-47	Introduction to environmental pollution, Types of environmental pollution. Cause, effect and control of air pollution. Major air pollutants and their sources and control measures. Cause, effect and control of water pollution. Thermal pollution and marine pollution. Basic concept of Eutrophication and biomagnification. Soil and noise pollution. Major soil pollutants and their impacts on human health.	Identify major types of pollution and their sources, exemplified by air pollution. Examine the causes, effects, and methods of controlling water pollution. Explore fundamental concepts like Eutrophication and bio-magnification. Investigate soil and noise pollution, including their impacts on the environment.	Lecture delivery with discussion	
		Environmental pollution (types and causes)	T-1 R-1	RW-26 RW-27 RW-28 RW-29 RW-30 AV-41 AV-42 AV-47	Introduction to environmental pollution, Types of environmental pollution. Cause, effect and control of air pollution. Major air pollutants and their sources and control measures. Cause, effect and control of water pollution. Thermal pollution and marine pollution. Basic concept of Eutrophication and biomagnification. Soil and noise pollution. Major soil pollutants and their impacts on human health.	Identify major types of pollution and their sources, exemplified by air pollution. Examine the causes, effects, and methods of controlling water pollution. Explore fundamental concepts like Eutrophication and bio-magnification. Investigate soil and noise pollution, including their impacts on the environment.	Lecture delivery with discussion	

Week 8	Lecture 15	Environmental pollution (effects and controls: air, water, soil, noise and radiation pollution)	T-1 R-1	RW-26 RW-27 RW-28 RW-29 RW-30 AV-41 AV-42 AV-47	Introduction to environmental pollution, Types of environmental pollution. Cause, effect and control of air pollution. Major air pollutants and their sources and control measures. Cause, effect and control of water pollution. Thermal pollution and marine pollution. Basic concept of Eutrophication and biomagnification. Soil and noise pollution. Major soil pollutants and their impacts on human health.	Identify major types of pollution and their sources, exemplified by air pollution. Examine the causes, effects, and methods of controlling water pollution. Explore fundamental concepts like Eutrophication and bio-magnification. Investigate soil and noise pollution, including their impacts on the environment.	Lecture delivery with discussion	
	Lecture 16	Environmental pollution (definition)	T-1 R-1	RW-26 RW-27 RW-28 RW-29 RW-30 AV-41 AV-42 AV-47	Introduction to environmental pollution, Types of environmental pollution. Cause, effect and control of air pollution. Major air pollutants and their sources and control measures. Cause, effect and control of water pollution. Thermal pollution and marine pollution. Basic concept of Eutrophication and biomagnification. Soil and noise pollution. Major soil pollutants and their impacts on human health.	Identify major types of pollution and their sources, exemplified by air pollution. Examine the causes, effects, and methods of controlling water pollution. Explore fundamental concepts like Eutrophication and bio-magnification. Investigate soil and noise pollution, including their impacts on the environment.	Lecture delivery with discussion	

Week 8	Lecture 16	Environmental pollution (types and causes)	T-1 R-1	RW-26 RW-27 RW-28 RW-29 RW-30 AV-41 AV-42 AV-47	Introduction to environmental pollution, Types of environmental pollution. Cause, effect and control of air pollution. Major air pollutants and their sources and control measures. Cause, effect and control of water pollution. Thermal pollution and marine pollution. Basic concept of Eutrophication and biomagnification. Soil and noise pollution. Major soil pollutants and their impacts on human health.	Identify major types of pollution and their sources, exemplified by air pollution. Examine the causes, effects, and methods of controlling water pollution. Explore fundamental concepts like Eutrophication and bio-magnification. Investigate soil and noise pollution, including their impacts on the environment.	Lecture delivery with discussion	
		Environmental pollution (effects and controls: air, water, soil, noise and radiation pollution)	T-1 R-1	RW-26 RW-27 RW-28 RW-29 RW-30 AV-41 AV-42 AV-47	Introduction to environmental pollution, Types of environmental pollution. Cause, effect and control of air pollution. Major air pollutants and their sources and control measures. Cause, effect and control of water pollution. Thermal pollution and marine pollution. Basic concept of Eutrophication and biomagnification. Soil and noise pollution. Major soil pollutants and their impacts on human health.	Identify major types of pollution and their sources, exemplified by air pollution. Examine the causes, effects, and methods of controlling water pollution. Explore fundamental concepts like Eutrophication and bio-magnification. Investigate soil and noise pollution, including their impacts on the environment.	Lecture delivery with discussion	

Week 9	Lecture 17	Environmental pollution (emerging pollutants)		RW-1 RW-3 RW-4 AV-44 AV-45	Ill effects of fireworks, Nuclear and radiation pollution. Cause, effect and control of radiation pollution. Solid waste management: control measures of urban and industrial waste. Pollution case study: Bhopal gas tragedy, Chernobyl accident, Ganga water pollution.	Examine the harmful effects of fireworks on the environment and public health. Explore the hazards associated with nuclear materials and their safe handling. Investigate the issues related to solid waste and learn about effective management procedures for waste disposal and recycling.	Lecture delivery with ppt, discussion and documentary clip.	
		Environmental pollution (fireworks and their ill-effects)		RW-1 RW-3 RW-4 AV-44 AV-45	Ill effects of fireworks, Nuclear and radiation pollution. Cause, effect and control of radiation pollution. Solid waste management: control measures of urban and industrial waste. Pollution case study: Bhopal gas tragedy, Chernobyl accident, Ganga water pollution.	Examine the harmful effects of fireworks on the environment and public health. Explore the hazards associated with nuclear materials and their safe handling. Investigate the issues related to solid waste and learn about effective management procedures for waste disposal and recycling.	Lecture delivery with ppt, discussion and documentary clip.	
		Environmental pollution (case studies on pollution)		RW-1 RW-3 RW-4 RW-31 AV-44 AV-45	Ill effects of fireworks, Nuclear and radiation pollution. Cause, effect and control of radiation pollution. Solid waste management: control measures of urban and industrial waste. Pollution case study: Bhopal gas tragedy, Chernobyl accident, Ganga water pollution.	Examine the harmful effects of fireworks on the environment and public health. Explore the hazards associated with nuclear materials and their safe handling. Investigate the issues related to solid waste and learn about effective management procedures for waste disposal and recycling.	Lecture delivery with ppt, discussion and documentary clip.	

Week 9	Lecture 18	Environmental pollution (climate change, global warming, ozone layer depletion)	T-1 R-1	RW-33 RW-34 AV-48 AV-49	Global climate change, Greenhouse effect and global warming. Effect of increase of Carbon dioxide and environmental effects of global warming, its control measure. Cause and mechanism of ozone layer depletion. Environmental effects of ozone layer depletion Causes and effects of acid rain.	Identify green house effect, global warming, and acid rain, and their impact on the environment. Analyze ozone layer depletion, its causes, and its environmental consequences.	Lecture delivery with necessary animations and videos.	
		Environmental pollution (acid rain and impacts on human communities and agriculture)	T-1 R-1	RW-33 RW-34 AV-48 AV-49	Global climate change, Greenhouse effect and global warming. Effect of increase of Carbon dioxide and environmental effects of global warming, its control measure. Cause and mechanism of ozone layer depletion. Environmental effects of ozone layer depletion Causes and effects of acid rain.	Identify green house effect, global warming, and acid rain, and their impact on the environment. Analyze ozone layer depletion, its causes, and its environmental consequences.	Lecture delivery with necessary animations and videos.	
Week 10	Lecture 19				Assignment - Field / industrial visit based			
	Lecture 20	Environmental policies and practices(environmental laws: Wildlife Protection Act)	T-1 R-1	RW-32 AV-50 AV-51 AV-52	Introduction to Environmental Legislation: Wildlife Protection Act, Water (Prevention and control of Pollution) Act, Forest Conservation Act.	Analyze environment legislation and various Environmental Laws.	Lecture delivery with discussion.	Conviction of Salman Khan under Section 51 of the Wildlife Protection Act in 2016 for killing BlackBuck, Odiase Austin Goubadia: Convicted under Section 51(1) of the Wildlife Protection Act

Week 10	Lecture 20	Environmental policies and practices(Water (Prevention and control of Pollution) Act)	T-1 R-1	RW-32 AV-50 AV-51 AV-52	Introduction to Environmental Legislation: Wildlife Protection Act, Water (Prevention and control of Pollution) Act, Forest Conservation Act.	Analyze environment legislation and various Environmental Laws.	Lecture delivery with discussion.	Conviction of Salman Khan under Section 51 of the Wildlife Protection Act in 2016 for killing BlackBuck, Odiase Austin Goubadia: Convicted under Section 51(1) of the Wildlife Protection Act
		Environmental policies and practices(Forest Conservation Act)	T-1 R-1	RW-32 AV-50 AV-51 AV-52	Introduction to Environmental Legislation: Wildlife Protection Act, Water (Prevention and control of Pollution) Act, Forest Conservation Act.	Analyze environment legislation and various Environmental Laws.	Lecture delivery with discussion.	Conviction of Salman Khan under Section 51 of the Wildlife Protection Act in 2016 for killing BlackBuck, Odiase Austin Goubadia: Convicted under Section 51(1) of the Wildlife Protection Act
Week 11	Lecture 21	Environmental policies and practices(Air (Prevention & Control of Pollution) Act)	T-1 R-1	AV-53 AV-54 AV-55 AV-58	Environmental legislation: Air (Prevention & Control of Pollution) Act, Environment (protection) Act, Environmental Impact Assessment, International agreements on environment protection: Montreal and Kyoto protocols	Understanding Air (Prevention & Control of Pollution) Act, Environment (protection) Act, Environmental Impact Assessment, International agreements on environment protection: Montreal and Kyoto protocols Investigate international agreements on environmental protection.	Lecture delivery with discussion.	

Week 11	Lecture 21	Environmental policies and practices(Environmental Protection Act)	T-1 R-1	AV-53 AV-54 AV-55 AV-58	Environmental legislation: Air (Prevention & Control of Pollution) Act, Environment (protection) Act, Environmental Impact Assessment, International agreements on environment protection: Montreal and Kyoto protocols	Understanding Air (Prevention & Control of Pollution) Act, Environment (protection) Act, Environmental Impact Assessment, International agreements on environment protection: Montreal and Kyoto protocols Investigate international agreements on environmental protection.	Lecture delivery with discussion.	
		Environmental policies and practices(Environmental Impact Assessment)	T-1 R-1	AV-53 AV-54 AV-55 AV-58	Environmental legislation: Air (Prevention & Control of Pollution) Act, Environment (protection) Act, Environmental Impact Assessment, International agreements on environment protection: Montreal and Kyoto protocols	Understanding Air (Prevention & Control of Pollution) Act, Environment (protection) Act, Environmental Impact Assessment, International agreements on environment protection: Montreal and Kyoto protocols Investigate international agreements on environmental protection.	Lecture delivery with discussion.	

Week 11	Lecture 21	Environmental policies and practices(International agreements: Montreal and Kyoto protocols)	T-1 R-1	AV-53 AV-54 AV-55 AV-58	Environmental legislation: Air (Prevention & Control of Pollution) Act, Environment (protection) Act, Environmental Impact Assessment, International agreements on environment protection: Montreal and Kyoto protocols	Understanding Air (Prevention & Control of Pollution) Act, Environment (protection) Act, Environmental Impact Assessment, International agreements on environment protection: Montreal and Kyoto protocols Investigate international agreements on environmental protection.	Lecture delivery with discussion.	
	Lecture 22	Environmental policies and practices(Air (Prevention & Control of Pollution) Act)	T-1 R-1	AV-53 AV-54 AV-55 AV-58	Environmental legislation: Air (Prevention & Control of Pollution) Act, Environment (protection) Act, Environmental Impact Assessment, International agreements on environment protection: Montreal and Kyoto protocols	Understanding Air (Prevention & Control of Pollution) Act, Environment (protection) Act, Environmental Impact Assessment, International agreements on environment protection: Montreal and Kyoto protocols Investigate international agreements on environmental protection.	Lecture delivery with discussion.	

Week 11	Lecture 22	Environmental policies and practices(Environmental Protection Act)	T-1 R-1	AV-53 AV-54 AV-55 AV-58	Environmental legislation: Air (Prevention & Control of Pollution) Act, Environment (protection) Act, Environmental Impact Assessment, International agreements on environment protection: Montreal and Kyoto protocols	Understanding Air (Prevention & Control of Pollution) Act, Environment (protection) Act, Environmental Impact Assessment, International agreements on environment protection: Montreal and Kyoto protocols Investigate international agreements on environmental protection.	Lecture delivery with discussion.	
		Environmental policies and practices(Environmental Impact Assessment)	T-1 R-1	AV-53 AV-54 AV-55 AV-58	Environmental legislation: Air (Prevention & Control of Pollution) Act, Environment (protection) Act, Environmental Impact Assessment, International agreements on environment protection: Montreal and Kyoto protocols	Understanding Air (Prevention & Control of Pollution) Act, Environment (protection) Act, Environmental Impact Assessment, International agreements on environment protection: Montreal and Kyoto protocols Investigate international agreements on environmental protection.	Lecture delivery with discussion.	

Week 11	Lecture 22	Environmental policies and practices(International agreements: Montreal and Kyoto protocols)	T-1 R-1	AV-53 AV-54 AV-55 AV-58	Environmental legislation: Air (Prevention & Control of Pollution) Act, Environment (protection) Act, Environmental Impact Assessment, International agreements on environment protection: Montreal and Kyoto protocols	Understanding Air (Prevention & Control of Pollution) Act, Environment (protection) Act, Environmental Impact Assessment, International agreements on environment protection: Montreal and Kyoto protocols Investigate international agreements on environmental protection.	Lecture delivery with discussion.	
Week 12	Lecture 23	Environmental policies and practices(Convention on Biological Diversity (CBD))	T-1 R-1	RW-2 AV-43 AV-46	Convention on Biological Diversity, Solid waste and its management: Control measures of urban and industrial waste. Role of tribal populations in policy making.	Understand the objectives, key principles and significance of the Convention on Biological Diversity (CBD). Analyze the role of countries in adopting and implementing CBD protocols to preserve biodiversity. Understand the classification of waste (organic, inorganic, hazardous, etc.). Explain the challenges associated with managing urban and industrial waste in rapidly growing cities and industries.	Lecture delivery with discussion.	Overflowing landfills, Fire outbreaks in landfills, hazardous waste disposal issues, or challenges in waste segregation in a particular city. India's commitment to the CBD through its National Biodiversity Action Plan and efforts to meet the Aichi Biodiversity Targets.

Week 12	Lecture 23	Environmental policies and practices(solid waste management: control measures of urban and industrial waste)	T-1 R-1	RW-2 AV-43 AV-46	Convention on Biological Diversity, Solid waste and its management: Control measures of urban and industrial waste. Role of tribal populations in policy making.	Understand the objectives, key principles and significance of the Convention on Biological Diversity (CBD). Analyze the role of countries in adopting and implementing CBD protocols to preserve biodiversity. Understand the classification of waste (organic, inorganic, hazardous, etc.). Explain the challenges associated with managing urban and industrial waste in rapidly growing cities and industries.	Lecture delivery with discussion.	Overflowing landfills, Fire outbreaks in landfills, hazardous waste disposal issues, or challenges in waste segregation in a particular city. India's commitment to the CBD through its National Biodiversity Action Plan and efforts to meet the Aichi Biodiversity Targets.
		Environmental policies and practices(role of tribal populations in policy making)	T-1 R-1	RW-2 AV-43 AV-46	Convention on Biological Diversity, Solid waste and its management: Control measures of urban and industrial waste. Role of tribal populations in policy making.	Understand the objectives, key principles and significance of the Convention on Biological Diversity (CBD). Analyze the role of countries in adopting and implementing CBD protocols to preserve biodiversity. Understand the classification of waste (organic, inorganic, hazardous, etc.). Explain the challenges associated with managing urban and industrial waste in rapidly growing cities and industries.	Lecture delivery with discussion.	Overflowing landfills, Fire outbreaks in landfills, hazardous waste disposal issues, or challenges in waste segregation in a particular city. India's commitment to the CBD through its National Biodiversity Action Plan and efforts to meet the Aichi Biodiversity Targets.
	Lecture 24				Test			

Week 13	Lecture 25	Human communities and environment(human population growth: impacts on environment)	T-1 R-1	RW-35 AV-2 AV-56 AV-57	Population growth: Birth rate, death rate, migration. Factors affecting variation of population: Causes and effects of population growth on human health, welfare and environment. Control of human population. Introduction to environmental movements, environmental ethics.	Recognize the causes and effects of population growth. Students will learn about Environmental ethics issues and possible solutions	Lecture delivery with discussion.	
		Human communities and environment(human health and welfare)	T-1 R-1	RW-35 AV-2	Population growth: Birth rate, death rate, migration. Factors affecting variation of population: Causes and effects of population growth on human health, welfare and environment. Control of human population. Introduction to environmental movements, environmental ethics.	Recognize the causes and effects of population growth. Students will learn about Environmental ethics issues and possible solutions	Lecture delivery with discussion.	Population clock of India and world.
		Human communities and environment(environmental movements: Chipko movement)	T-1 R-1	RW-35 AV-2	Population growth: Birth rate, death rate, migration. Factors affecting variation of population: Causes and effects of population growth on human health, welfare and environment. Control of human population. Introduction to environmental movements, environmental ethics.	Recognize the causes and effects of population growth. Students will learn about Environmental ethics issues and possible solutions	Lecture delivery with discussion.	Population clock of India and world.

Week 13	Lecture 25	Human communities and environment(environmental ethics)	T-1 R-1	RW-35 AV-2	Population growth: Birth rate, death rate, migration. Factors affecting variation of population: Causes and effects of population growth on human health, welfare and environment. Control of human population. Introduction to environmental movements, environmental ethics.	Recognize the causes and effects of population growth. Students will learn about Environmental ethics issues and possible solutions	Lecture delivery with discussion.	Population clock of India and world.
	Lecture 26	Human communities and environment(Silent valley movement)	T-1 R-1	OR-2 OR-3 RW-5 RW-6	Human Communities and the Environment (environmental movements: Chipko movement, Apikko movement, silent valley movement, Bishnois of Rajasthan, Narmada Bachao Andolan,, Jungle bachao andolan, Tehri Dam conflict) role of Indian and other religions and cultures in environmental conservation and public awareness	Students will learn about various Environmental movements and environmental communication. Recognize role of individual and communities in environmental protection.	Lecture delivery with the help of ppt, clips, discussion and documentary clips.	

Week 13	Lecture 26	Human communities and environment(Appiko movement)	T-1 R-1	OR-2 OR-3 RW-5 RW-6	Human Communities and the Environment (environmental movements: Chipko movement, Apikko movement, silent valley movement, Bishnois of Rajasthan, Narmada Bachao Andolan,, Jungle bachao andolan, Tehri Dam conflict) role of Indian and other religions and cultures in environmental conservation and public awareness	Students will learn about various Environmental movements and environmental communication. Recognize role of individual and communities in environmental protection.	Lecture delivery with the help of ppt, clips, discussion and documentary clips.	
		Human communities and environment(Jungle bachao andolan)	T-1 R-1	OR-2 OR-3 RW-5 RW-6	Human Communities and the Environment (environmental movements: Chipko movement, Apikko movement, silent valley movement, Bishnois of Rajasthan, Narmada Bachao Andolan,, Jungle bachao andolan, Tehri Dam conflict) role of Indian and other religions and cultures in environmental conservation and public awareness	Students will learn about various Environmental movements and environmental communication. Recognize role of individual and communities in environmental protection.	Lecture delivery with the help of ppt, clips, discussion and documentary clips.	

Week 13	Lecture 26	Human communities and environment(Narmada Bachao Andolan)	T-1 R-1	OR-2 OR-3 RW-5 RW-6	Human Communities and the Environment (environmental movements: Chipko movement, Apikko movement, silent valley movement, Bishnois of Rajasthan, Narmada Bachao Andolan,, Jungle bachao andolan, Tehri Dam conflict) role of Indian and other religions and cultures in environmental conservation and public awareness	Students will learn about various Environmental movements and environmental communication. Recognize role of individual and communities in environmental protection.	Lecture delivery with the help of ppt, clips, discussion and documentary clips.	
		Human communities and environment(Tehri Dam conflict)	T-1 R-1	OR-2 OR-3 RW-5 RW-6	Human Communities and the Environment (environmental movements: Chipko movement, Apikko movement, silent valley movement, Bishnois of Rajasthan, Narmada Bachao Andolan,, Jungle bachao andolan, Tehri Dam conflict) role of Indian and other religions and cultures in environmental conservation and public awareness	Students will learn about various Environmental movements and environmental communication. Recognize role of individual and communities in environmental protection.	Lecture delivery with the help of ppt, clips, discussion and documentary clips.	

Week 13	Lecture 26	Human communities and environment(Bishnois of Rajasthan)	T-1 R-1	OR-2 OR-3 RW-5 RW-6	Human Communities and the Environment (environmental movements: Chipko movement, Apikko movement, silent valley movement, Bishnois of Rajasthan, Narmada Bachao Andolan,, Jungle bachao andolan, Tehri Dam conflict) role of Indian and other religions and cultures in environmental conservation and public awareness	Students will learn about various Environmental movements and environmental communication. Recognize role of individual and communities in environmental protection.	Lecture delivery with the help of ppt, clips, discussion and documentary clips.	
		Human communities and environment(role of Indian and other religions and cultures in environmental conservation)	T-1 R-1	OR-2 OR-3 RW-5 RW-6	Human Communities and the Environment (environmental movements: Chipko movement, Apikko movement, silent valley movement, Bishnois of Rajasthan, Narmada Bachao Andolan,, Jungle bachao andolan, Tehri Dam conflict) role of Indian and other religions and cultures in environmental conservation and public awareness	Students will learn about various Environmental movements and environmental communication. Recognize role of individual and communities in environmental protection.	Lecture delivery with the help of ppt, clips, discussion and documentary clips.	

Week 13	Lecture 26	Human communities and environment(public awareness)	T-1 R-1	OR-2 OR-3 RW-5 RW-6	Human Communities and the Environment (environmental movements: Chipko movement, Apikko movement, silent valley movement, Bishnois of Rajasthan, Narmada Bachao Andolan,, Jungle bachao andolan, Tehri Dam conflict) role of Indian and other religions and cultures in environmental conservation and public awareness.	Students will learn about various Environmental movements and environmental communication. Recognize role of individual and communities in environmental protection.	Lecture delivery with the help of ppt, clips, discussion and documentary clips.	
Week 14	Lecture 27	Human communities and environment(disaster management)	T-1 R-1	RW-36 AV-15 AV-16	Disaster and disaster management : Types of disaster, cause, effects and management of floods, earthquake, cyclones and landslides.	Explore various aspects of disasters and disaster management.	Lecture delivery with discussion.	Recent disaster happening and their management across India and Globe
		Human communities and environment(floods, droughts, earthquake)	T-1 R-1	RW-36 AV-15 AV-16	Disaster and disaster management : Types of disaster, cause, effects and management of floods, earthquake, cyclones and landslides.	Explore various aspects of disasters and disaster management.	Lecture delivery with discussion.	Recent disaster happening and their management across India and Globe
		Human communities and environment(landslides and cyclones)	T-1 R-1	RW-36 AV-15 AV-16	Disaster and disaster management : Types of disaster, cause, effects and management of floods, earthquake, cyclones and landslides.	Explore various aspects of disasters and disaster management.	Lecture delivery with discussion.	Recent disaster happening and their management across India and Globe
		SPILL OVER						
Week 14	Lecture 28				Spill Over			
Week 15	Lecture 29				Spill Over			
	Lecture 30				Spill Over			

Scheme for CA:

CA Category of this Course Code is:A0203 (2 best out of 3)

Component	Weightage (%)	Mapped CO(s)
Test	50	CO1, CO2, CO3, CO4
Assignment - Field / industrial visit based	50	CO1, CO2, CO3, CO4

Details of Academic Task(s)

Academic Task	Objective	Detail of Academic Task	Nature of Academic Task (group/individuals)	Academic Task Mode	Marks	Allottment / submission Week
Assignment - Field / industrial visit based	To make the student understand about environmental problems and propose some solution by engaging them into various field work based assignment and individual participation	To submit elaborative term paper report and supporting information related to the fieldwork done. The marks distribution will be as follows Total -100 marks. Final term paper report-50 marks Supporting information- 50 marks.	Group	Offline	100	2 / 10
Test	To check the student learning and concept understanding about environmental problems and their solutions	Total-30 MCQs from Unit-I to Unit-VI. Five questions from each Unit. Each question will carry one mark.	Individual	Online	30	10 / 12

List of suggested topics for term paper[at least 15] (Student to spend about 15 hrs on any one specified term paper)

Sr. No.	Topic
1	Importance of solar energy as renewable source of energy and projects to strengthen solar energy network in India.
2	Water wastage at public places: Scope of technology based control of water wastage.
3	To identify major health problems suffered by human population by the use of electronic gadgets (mobiles/laptops).

An instruction plan is only a tentative plan. The teacher may make some changes in his/her teaching plan. The students are advised to use syllabus for preparation of all examinations. The students are expected to keep themselves updated on the contemporary issues related to the course. Upto 20% of the questions in any examination/Academic tasks can be asked from such issues even if not explicitly mentioned in the instruction plan.

4	Plastic waste: Sources, disposal/management and availability and use of E-crusher at nearby public places.
5	Electronic wastes as future environmental problem of multinationals and Institutions: Practical solutions.
6	Rainwater harvesting: Survey on applicability and benefits.
7	Recycling of metals from electronic gadget: An approach to control metal pollution.
8	Smart watering: automated sprinkler and their role in water conservation (applicability in and around your place).
9	Sources of noise pollution in your area and practical solutions to control them.
10	Solar-power plant: awareness on applicability among students about benefits of solar energy.
11	An analysis on recycling of metals from electronic gadget: An approach to control metal pollution and to understand health problems to worker.
12	Groundwater pollution due to leaching from e-wastes components: Case study.
13	Survey on availability of online system for handling and disposal of produced waste from public places.
14	Identification of electrical energy wastage due to electronic gadgets : Practical solutions to reduce wastage.
15	Solid waste and its management: Problem with the landfill sites and environmental pollution.